AI.312

Social Robotics

COURSE OVERVIEW

Robots in society: social norms, long-term interaction, ethics, and robots as companions and collaborators.



ILLUSTRATIVE HERO IMAGERY

1

SESSION

60 MINUTES

ORIENTATION: ROBOTS IN SOCIETY

KEY TOPICS

- What is social robotics
- Social robots vs. industrial robots
- Roles: companion, tutor, caregiver
- Course roadmap

TEACHING FOCUS

Design robots for social worlds.

KEY TAKEAWAY Social robots relate, not just perform.

2

SESSION

60 MINUTES

SOCIAL COGNITION

KEY TOPICS

- Joint attention
- Turn-taking
- Empathy and rapport
- Social intelligence

TEACHING FOCUS

Read and respond to social cues.

KEY TAKEAWAY Social skill is interaction skill.

3

SESSION

60 MINUTES

LONG-TERM INTERACTION

KEY TOPICS

- Relationship building
- Adaptation over time
- User modeling
- Engagement

TEACHING FOCUS

Sustain interaction across encounters.

KEY TAKEAWAY Trust grows over repeated encounters.

4

SESSION

60 MINUTES

ETHICS OF SOCIAL ROBOTS

KEY TOPICS

- Attachment and dependency
- Deception and authenticity
- Privacy in the home
- Vulnerable populations

TEACHING FOCUS

Protect users in intimate settings.

KEY TAKEAWAY Social robots must respect dignity.

SOCIAL ROBOTICS

5

SESSION

60 MINUTES

APPLICATIONS

KEY TOPICS

- Education robots
- Healthcare and elder care
- Retail and hospitality
- Therapeutic robots

TEACHING FOCUS

Deploy robots in social institutions.

KEY TAKEAWAY Social robots serve people in context.

SOCIAL ROBOTICS

6

SESSION

60 MINUTES

SOCIAL ROBOTICS & REVIEW

KEY TOPICS

- Social robot demo
- Course recap and assessment
- Toward evolutionary robotics (AI.401)
- Ethical reflection

TEACHING FOCUS

Evaluate a robot's social impact.

KEY TAKEAWAY A social robot is measured by human well-being.